Quiz 14-16 - Ideal gas and Thermodynamics

Name: Class:

The number of marks for each question or part question is shown in brackets [ ]. Total mark for this section: 40.

1.

a.

i. State the magnitude and unit of absolute zero on the thermodynamic temperature scale.

.....[1]

ii. Explain why temperature measured using a laboratory liquid-in-glass thermometer does not give a measurement of thermodynamic temperature.

.....[1]

b. Fig. 2.1 shows a simplified diagram of a type of thermometer called a platinum resistance thermometer.

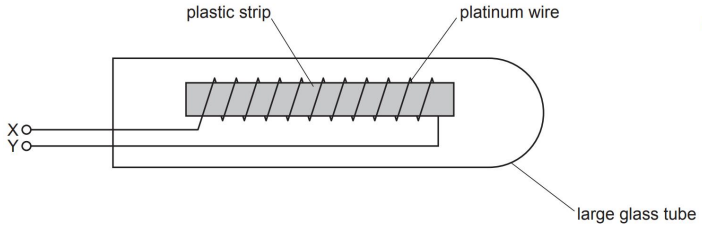


Fig. 2.1

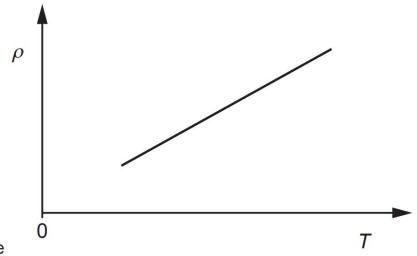


Fig. 2.2

The glass tube is immersed in the environment for which the temperature is to be determined. The resistance between the terminals X and Y is measured.

Fig. 2.2 shows the variation of the resistivity ρ of platinum with thermodynamic temperature T.

i. Explain how Fig. 2.2 shows that platinum is a suitable metal for use in a resistance thermometer.

.....

.....[2]

ii. Suggest a reason why a platinum resistance thermometer is not suitable for measuring a rapidly changing temperature.

.....[1]

iii. Suggest a type of thermometer that is suitable for measuring a rapidly changing temperature.

.....[1]

c. A negative temperature coefficient thermistor may be used as a type of resistance thermometer. State one way in which the variation with temperature of the resistance of a thermistor differs from that of a platinum wire

.....[1]

[Total: 7]

2.

a. Define specific heat capacity.

.....

.....[2]

b. A sealed container of fixed volume V contains N molecules, each of mass m, of an ideal gas at pressure p.

i. State an expression, in terms of V, N, p and the Boltzmann constant k, for the thermodynamic temperature T of the gas.

.....[1]

ii. Show that the mean translational kinetic energy E_K of a molecule of the gas is given by [2]

$$E_K = \frac{3}{2}kT.$$

iii. Explain why the internal energy of the gas is equal to the total kinetic energy of the molecules.

.....

.....[2]

c. The gas in (b) is supplied with thermal energy Q.

i. Explain, with reference to the first law of thermodynamics, why the increase in internal energy of the gas is Q.

.....

.....[2]

ii. Use the expression in (b)(ii) and the information in (c)(i) to show that the specific heat capacity c of the gas is given by [2]

$$c = \frac{3k}{2m}.$$

d. The container in (b) is now replaced with one that does not have a fixed volume. Instead, the gas is able to expand, so that the pressure of the gas remains constant as thermal energy is supplied. Suggest, with a reason, how the specific heat capacity of the gas would now compare with the value in (c)(ii).

.....

.....[2]

[Total: 13]

3.

a. State four of the basic assumptions of the kinetic theory of gases.

1.....

2.....

3.....

4.....[4]

b. Explain how molecular movement causes the pressure exerted by a gas.

.....

.....

.....[3]

c. Fig. 4.1 shows the variation with thermodynamic temperature T of the mean-square speeds $<c^2>$ for two gases X and Y.

Fig. 4.2 shows the variation with T of the product pV for samples of the two gases, where p is the pressure of the gas and V is the volume of the gas.

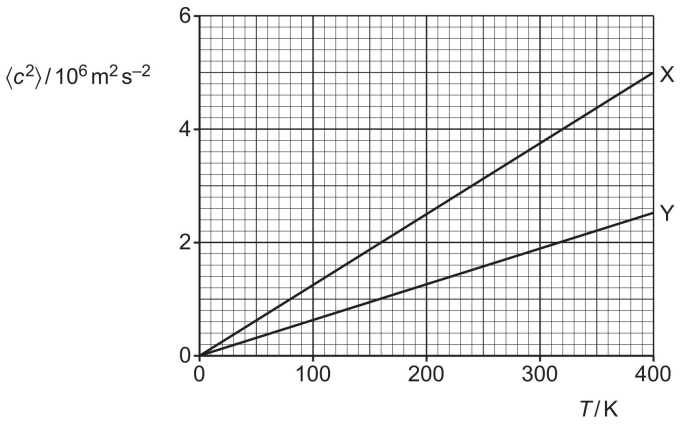


Fig. 4.1

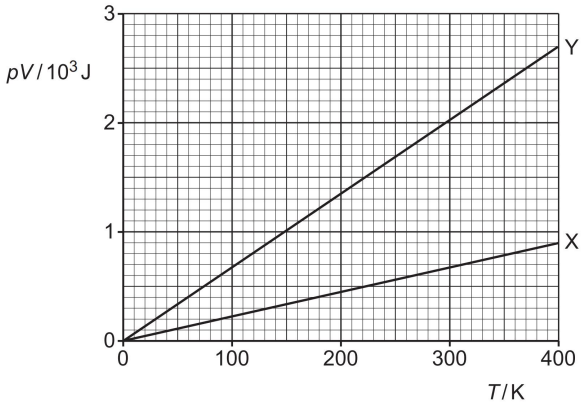


Fig. 4.2

State three conclusions about the gases and their samples that may be drawn from Fig. 4.1 and Fig. 4.2. The conclusions may be qualitative or quantitative. Use the space below for any working that you need.

1.....

2.....

3.....[3]

[Total: 10]

4. An ideal gas has amount of substance n.

The gas is initially in state X, with pressure 2p and volume V.

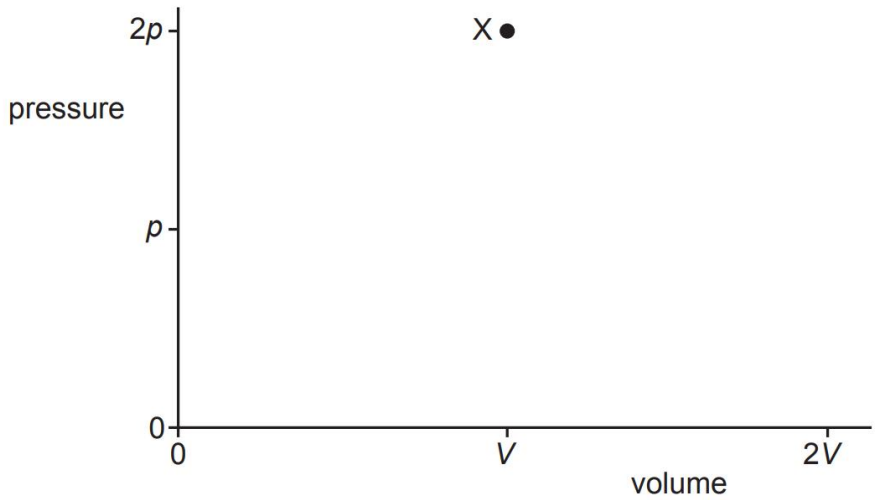
The gas is cooled at constant volume to state Y, with pressure p.

The gas is then heated at constant pressure to state Z, with volume 2V.

Finally, the gas returns at constant temperature to state X.

a. Determine an expression for the temperature T of the gas in state X, in terms of n, p and V. Identify any other symbols that you use. [2]

b. On Fig., sketch the variation with volume of pressure for the gas as the gas undergoes the three changes. The state X is labelled. Label states Y and Z. [3]



c. During the change of state from Y to Z, the increase in internal energy of the gas is U.

During the change of state from Z to X, the work done on the gas is W.

Complete Table to indicate, for each of the three changes of state, the increase in internal energy of the gas, the thermal energy transferred to the gas and the work done on the gas, in terms of p, V, U and W. [5]

change	increase in internal energy of gas	thermal energy transferred to gas	work done on gas
X to Y			
Y to Z	$+U$		
Z to X			$+W$

[Total: 10]